How Much of Earnings Information is Factor Compensation? Evidence from a Systematic Decomposition of Abnormal Returns

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Abstract

Earnings surprises generate well-documented abnormal returns, but what fraction of this apparent information content reflects genuine news versus compensation for systematic risk factors? We address this question through a comprehensive decomposition using five abnormal return models—from Ball and Brown's (1968) simple market adjustment through the Fama–French–Carhart four-factor specification—applied to 2,358 U.S. firms with December 2023 fiscal year-ends. Testing 20 specifications that combine five models, two surprise measures (analyst-based and time-series), and two event definitions, we document a striking attenuation pattern: under time-series surprises, the good-news/bad-news spread declines from 12.98 percentage points to 0.16 percentage points as the model moves from simple market adjustment to four-factor adjustment—a reduction of 99%. Even analyst-based surprises attenuate by 65%, from 9.00 to 3.15 percentage points. Analyst-based surprises survive four-factor adjustment; time-series surprises do not. Post-earnings announcement drift approximately doubles the spread by month +12 and concentrates overwhelmingly in small-capitalization stocks (14.25 percentage points of drift versus 3.03 for large caps), consistent with limited-attention theories. Cross-sectional regressions confirm the predictive power of analyst surprises after controlling for size, book-to-market, momentum, analyst coverage, earnings volatility, leverage, and industry fixed effects. These findings carry direct implications for portfolio construction around earnings events.

1 Introduction

Earnings announcements are among the most actively traded corporate events in financial markets. Institutional and retail investors alike build strategies around earnings surprises, and a large industry of analyst forecasting exists to anticipate these events. The notion that accounting earnings contain value-relevant information—that stock prices respond systematically to earnings surprises—has been established since the seminal work of Ball and Brown [1968]. Yet a fundamental question remains underexplored: of the measured abnormal return surrounding earnings announcements, how much represents genuine information content and how much is compensation for exposure to systematic risk factors?

This question has direct implications for both academic research and investment practice. If the earnings surprise spread is largely attributable to factor loadings, then earnings-based trading strategies may not generate true alpha but instead harvest risk premia that investors could access more efficiently through factor portfolios. Conversely, if a meaningful residual survives comprehensive risk adjustment, this implies that earnings announcements convey idiosyncratic information that the market fails to fully and immediately incorporate—a violation of semi-strong form efficiency as defined by Fama [1970].

Ball and Brown [1968] established the information content of earnings in their landmark study of 261 New York Stock Exchange firms over 1957–1965. They introduced the Abnormal Performance Index (API) methodology and demonstrated that firms with positive unexpected earnings ("good news") systematically outperform those with negative surprises ("bad news"), with the two groups diverging steadily over the twelve months preceding the earnings announcement. Their key finding—that approximately 85–90% of annual earnings information is anticipated by the market before the formal announcement—has been confirmed in numerous subsequent studies and remains one of the most robust empirical regularities in financial economics [Kothari, 2001].

Despite extensive research on earnings information content over the past six decades, a systematic decomposition of the Ball and Brown spread across a comprehensive grid of

abnormal return models has not been conducted. Individual studies have applied specific models—the market model [Brown and Warner, 1985], the Fama–French three-factor model [Fama and French, 1993], or the Carhart four-factor extension [Carhart, 1997]—but no study has simultaneously varied the abnormal return model, the earnings surprise measure, and the event date definition to ask: how much of the spread survives progressively tighter risk adjustment? The answer to this question speaks directly to the joint hypothesis problem articulated by Fama [1970]: any test of market efficiency is simultaneously a test of the asset pricing model used to define "normal" returns.

We fill this gap by testing 20 event study specifications that arise from the intersection of five abnormal return models (simple equal-weighted market-adjusted, simple value-weighted market-adjusted, regression-based market model with each benchmark, and the Fama–French–Carhart four-factor model), two earnings surprise measures (analyst-based Standardized Unexpected Earnings and Ball and Brown's original time-series measure), and two event date definitions (fiscal year-end and actual announcement date). We apply these 20 specifications to a sample of 2,358 U.S. common stocks with December 2023 fiscal year-ends and I/B/E/S analyst coverage.

Our results reveal a striking and systematic attenuation pattern. Using Ball and Brown's exact methodology—simple equal-weighted market-adjusted returns with time-series earnings surprises measured at the fiscal year-end—we obtain a 12.98 percentage point spread between good-news and bad-news portfolios, closely matching the original finding of 15–20 percentage points. Moving along this specification while progressively tightening risk adjustment, the spread declines to 12.20 percentage points under simple value-weighted adjustment, approximately 3.6 percentage points under the regression-based market model, and just 0.16 percentage points under the Fama–French–Carhart four-factor model—a reduction of nearly 99%. Using analyst-based surprises (SUE) measured at the announcement date, the same progression yields 9.00, 7.60, 5.23, and 3.15 percentage points—a 65% reduction. Across both paths, the pattern is the same: the bulk of what has traditionally been measured as

"earnings information content" is in fact compensation for exposure to market, size, value, and momentum factors.

An important asymmetry emerges between our two surprise measures. Under four-factor adjustment, analyst-based surprises retain a statistically significant spread of 3.15 percentage points (bootstrap 95% CI: $[0.55, 5.80]$). Time-series surprises, by contrast, yield a spread of -0.26 percentage points (95% CI: $[-2.57, 1.90]$), indistinguishable from zero. This divergence is consistent with the findings of Livnat and Mendenhall [2006], who documented stronger post-earnings announcement drift for analyst-based surprises, and suggests that analyst consensus forecasts incorporate idiosyncratic information beyond what is captured by systematic risk factors.

Extending our event window to $[-12, +12]$ months relative to the announcement date, we document substantial post-earnings announcement drift (PEAD) consistent with Bernard and Thomas [1989, 1990]. The good-news/bad-news spread approximately doubles from the announcement date to month $+12$ across all model specifications, and this drift survives four-factor adjustment. Decomposing by firm size, we find that PEAD concentrates overwhelmingly in small-capitalization stocks: the post-announcement drift is 14.25 percentage points for small caps versus only 3.03 percentage points for large caps—a nearly fivefold difference. This size-conditional pattern is consistent with the limited-attention hypothesis of Hong et al. [2000], who argued that information diffuses slowly into the prices of firms with lower analyst coverage and institutional ownership.

Our study makes three contributions. First, we provide a systematic decomposition of the Ball and Brown earnings information spread across the full spectrum of standard abnormal return models, quantifying for the first time the fraction attributable to factor compensation. Second, we demonstrate that the choice between analyst-based and time-series surprise measures has first-order consequences under factor adjustment—analyst forecasts contain genuine idiosyncratic information that time-series models do not—with implications for how researchers and practitioners measure earnings surprises. Third, we document size-

conditional PEAD that is consistent with limited-attention theories and carries direct implications for portfolio construction: earnings-based alpha concentrates in small-cap stocks and accrues primarily in the first three months following announcement.

The remainder of this paper is organized as follows. Section 2 reviews the related literature and develops our hypotheses. Section 3 describes the data and methodology. Section 4 presents the empirical results. Section 5 discusses implications for market efficiency and portfolio management. Section 6 concludes.

2 Literature and hypotheses

Our study connects to three streams of literature: the information content of earnings, the sensitivity of event study results to model specification, and the post-earnings announcement drift anomaly. We review each and develop testable hypotheses.

2.1 Information content of earnings

The foundational result in this literature is Ball and Brown [1968], who demonstrated that accounting earnings possess information content by showing that abnormal stock returns are systematically associated with the sign of unexpected earnings changes. Using a sample of 261 NYSE firms over 1957–1965, they computed monthly abnormal returns as the difference between individual stock returns and market portfolio returns (the simple market-adjusted model) and constructed the Abnormal Performance Index to track cumulative abnormal performance for good-news and bad-news portfolios. Their finding that the two portfolios diverge by approximately 15–20 percentage points over the twelve months surrounding the earnings announcement, with most of this divergence occurring before the formal release, established the information content paradigm that has shaped financial accounting research for over half a century.

Contemporaneously, Beaver [1968] examined trading volume and return variance around

annual earnings announcement dates and found significant increases in both, providing complementary evidence for the information content of earnings from a market microstructure perspective. Together, the Ball and Brown and Beaver studies launched the empirical capital markets research program in accounting.

Subsequent work has confirmed and extended these results. Kothari [2001], in a comprehensive survey of capital markets research in accounting, documented the robustness of the earnings–return association across different sample periods, countries, and methodological approaches. The earnings response coefficient literature, beginning with Ball and Brown [1968] and formalized by subsequent studies, established that the magnitude of the stock price response is increasing in the size of the earnings surprise and the persistence of earnings, and decreasing in the riskiness of the firm.

Foster et al. [1984] compared alternative measures of unexpected earnings and their associations with abnormal returns. They introduced the Standardized Unexpected Earnings (SUE) measure computed from time-series models and demonstrated that the magnitude of measured abnormal returns varies substantially with the choice of expectation model. This finding motivates our systematic comparison of analyst-based and time-series surprise measures, as the choice of surprise definition can alter both the classification of firms into good-news and bad-news groups and the measured spread between them.

More recently, Livnat and Mendenhall [2006] directly compared post-earnings announcement drift for surprises computed from analyst consensus forecasts versus time-series statistical models. They found that analyst-based SUE generates significantly stronger drift, suggesting that analyst forecasts capture aspects of market expectations that time-series models miss. Our decomposition extends this comparison by testing both measures across five abnormal return models and two event definitions, providing a comprehensive map of how surprise measurement interacts with risk adjustment.

Based on this literature, we formulate our first hypothesis:

H1 (Information Content Persistence): *The systematic association between earnings*

surprises and abnormal stock returns documented by Ball and Brown (1968) persists in modern data, with good-news portfolios outperforming bad-news portfolios across model specifications.

2.2 Model specification and the joint hypothesis problem

Fama [1970] defined three forms of market efficiency—weak, semi-strong, and strong—and articulated the joint hypothesis problem: any test of market efficiency is simultaneously a test of the equilibrium model used to define expected returns. Rejecting efficiency may simply mean the asset pricing model is misspecified. This insight implies that the measured "information content" of earnings depends critically on the benchmark model used to compute abnormal returns.

Brown and Warner [1985] provided the methodological foundation for event studies using daily stock returns, establishing the 250-day estimation window with a gap before the event as standard practice. They compared the performance of three models—mean-adjusted returns, market-adjusted returns, and the ordinary least squares (OLS) market model—and found that simpler models often perform comparably to the market model in random samples, though the market model offers better specification when the event is clustered in calendar time. MacKinlay [1997] subsequently presented a comprehensive modern treatment of event study methodology, including the statistical properties of test statistics under different return-generating processes.

The introduction of multifactor models substantially altered the landscape. Fama and French [1993] demonstrated that much of the cross-sectional variation in expected stock returns can be explained by three factors: the market portfolio, a size factor (SMB), and a value factor (HML). Carhart [1997] extended this framework by adding a momentum factor (UMD), motivated by the findings of Jegadeesh and Titman [1993] that buying past winners and selling past losers generates significant profits over 3–12 month horizons. The four-factor model has become the standard benchmark for evaluating abnormal performance in

the post-1993 literature.

The implication for earnings event studies is that a portion of what appears to be earnings-related abnormal performance under simple models may reflect systematic factor exposures. For example, firms with positive earnings surprises may tend to be past winners (loading positively on momentum) or small-growth firms (loading on size and value factors). If so, the earnings surprise spread measured under simple market adjustment would overstate the genuine information content relative to a four-factor benchmark. The magnitude of this overstatement has not been systematically quantified.

Fama [1998] argued that many apparent long-run return anomalies, including PEAD, are sensitive to the methodology used and may reflect problems with risk adjustment rather than true market inefficiency. While his argument focused primarily on long-horizon studies, the logic applies equally to the event window itself: if the measured abnormal return depends on model choice, the "information content" is partially an artifact of model specification. Our 20-specification grid directly addresses this concern by documenting precisely how the earnings spread changes as we move from the simplest to the most comprehensive risk adjustment.

This motivates our second hypothesis:

H2 (Attenuation with Risk Adjustment): *The measured spread between good-news and bad-news earnings portfolios attenuates monotonically as the abnormal return model incorporates additional systematic risk factors, from simple market adjustment through the Fama–French–Carhart four-factor specification.*

2.3 Post-earnings announcement drift and limited attention

Bernard and Thomas [1989] documented that stock prices continue to drift in the direction of the earnings surprise for approximately 60 trading days following the announcement—a phenomenon they termed post-earnings announcement drift (PEAD). In a companion paper, Bernard and Thomas [1990] provided evidence that the drift reflects the market's failure

to fully incorporate the implications of current earnings for future earnings, specifically the predictable autocorrelation structure of quarterly earnings changes. PEAD is widely regarded as one of the most robust anomalies in the asset pricing literature, having survived decades of out-of-sample tests and corrections for risk factors.

The persistence of PEAD poses a challenge to semi-strong form efficiency. Several explanations have been advanced. Fama [1998] suggested that the drift may reflect risk mismeasurement rather than inefficiency. However, Bernard and Thomas [1989] showed that PEAD generates abnormal returns even after controlling for systematic risk, and our own results confirm that the drift survives Fama–French–Carhart four-factor adjustment.

A prominent behavioral explanation centers on limited investor attention. Hong et al. [2000] developed a model in which information diffuses gradually across the investing population and provided empirical evidence that momentum profits—which share features with PEAD—are more pronounced for stocks with low analyst coverage. Their logic applies directly to PEAD: if the market's initial response to earnings news is incomplete because not all investors process the information immediately, the resulting drift should be stronger for firms that attract less investor attention, which are typically smaller firms with fewer analysts.

Banz [1981] first documented the size effect—that small-capitalization stocks earn higher average returns than large-capitalization stocks, even after adjusting for market risk. While Fama and French [1993] proposed that size captures a compensated risk factor, the interaction between size and PEAD may reflect a distinct channel: smaller firms have lower institutional ownership, less analyst coverage, and less media attention, all of which slow the diffusion of earnings information into prices.

Combining these insights, we formulate our third hypothesis:

H3 (Size-Conditional PEAD): *Post-earnings announcement drift is stronger for small-capitalization firms than for large-capitalization firms, consistent with limited-attention explanations for the market's incomplete initial response to earnings news.*

3 Data and methodology

3.1 Sample construction

Our sample comprises U.S. common stocks with I/B/E/S analyst coverage and December fiscal year-ends for fiscal year 2023. We focus on FY2023 because earnings for this fiscal year were announced in the first quarter of 2024, and CRSP return data are available through December 2024, providing a complete event window including post-announcement drift.

We begin with all firms in the I/B/E/S Summary Statistics file (`ibes.statsum_epsus`) that have annual EPS forecasts (forecast period indicator `fpi = '1'`), a December fiscal period end date, a non-missing actual EPS value, at least three contributing analysts, and a U.S. firm indicator. For firms with multiple consensus records for the same fiscal period, we retain the most recent statistical period. This yields 3,187 firm-level observations.

We link I/B/E/S tickers to CRSP permanent security identifiers (PERMNOs) using the WRDS I/B/E/S–CRSP linking table (`wrdsapps.ibcrsphist`). When multiple links exist for a given ticker, we retain the link with the highest quality score. We require that the event date (either fiscal year-end or announcement date) falls within the link validity period. We further require a share price of at least $5.00 as of the last available CRSP trading date on or before the fiscal year-end, excluding penny stocks that may generate extreme abnormal returns due to microstructure effects. The final sample contains 2,358 firms with valid SUE classifications and 1,900 firms with valid TSUE classifications. Table 1 summarizes the sample construction.

3.2 Earnings surprise measures

We employ two measures of unexpected earnings that correspond to different information sets.

Table 1: Sample construction

Step	Firms
FY2023 December FYE with actual EPS and ≥ 3 analysts (I/B/E/S)	3,187
After computing TSUE (requires ≥ 3 years of historical EPS)	2,903
After linking to CRSP (valid PERMNO and link period)	2,358
With valid SUE classification	2,358
With valid TSUE classification	1,900
After model estimation (minimum 12 estimation months)	
Fiscal year-end event panel	2,298
Announcement date event panel	2,306

This table reports the number of firms at each stage of sample construction. The initial sample includes all U.S. firms in I/B/E/S with December fiscal year-ends, annual EPS actuals, and at least three contributing analysts for FY2023. TSUE requires a minimum of three years of historical EPS data (2015–2022). CRSP linking uses the `wrdsapps.ibcrsphist` table with a \$5 minimum price filter. Model estimation requires at least 12 monthly return observations in the estimation window (months -24 to -4 relative to the event).

Analyst-based Standardized Unexpected Earnings (SUE). Following Foster et al. [1984] and subsequent literature, we compute

$$\text{SUE}_i = \frac{\text{Actual}_i - \text{Median Consensus}_i}{\sigma_i}, \tag{1}$$

where Actual_i is the reported EPS, $\text{Median Consensus}_i$ is the median analyst forecast from the most recent consensus period, and σ_i is the cross-sectional standard deviation of individual analyst forecasts. We winsorize SUE at the 1st and 99th percentiles to limit the influence of outliers. Firms with SUE > 0 are classified as good news; firms with SUE < 0 as bad news. In the initial I/B/E/S sample of 3,187 firms, 1,768 are classified as good news and 1,273 as bad news, with the remainder having SUE exactly equal to zero. These proportions are broadly preserved after linking to CRSP and filtering, yielding 2,358 firms with valid SUE classifications in the final sample.

Time-Series Unexpected Earnings (TSUE). Following the original approach of Ball and Brown [1968], we estimate a time-series regression of firm-level earnings changes on

market-wide earnings changes:

$$\Delta\text{EPS}_{i,t} = \alpha_i + \beta_i \cdot \Delta\overline{\text{EPS}}_t + \varepsilon_{i,t}, \tag{2}$$

where $\Delta\text{EPS}_{i,t} = \text{EPS}_{i,t} - \text{EPS}_{i,t-1}$ is the year-over-year change in actual EPS for firm i, and $\Delta\overline{\text{EPS}}_t$ is the cross-sectional mean of ΔEPS across all firms in year t. We estimate this regression over fiscal years 2015–2022 (requiring at least three annual observations) and define the time-series unexpected earnings for FY2023 as the residual:

$$\text{TSUE}_{i,2023} = \Delta\text{EPS}_{i,2023} - (\hat{\alpha}_i + \hat{\beta}_i \cdot \Delta\overline{\text{EPS}}_{2023}). \tag{3}$$

Under the TSUE classification, 1,265 of the 2,903 firms with sufficient historical data are classified as good news and 1,638 as bad news. After linking to CRSP, 1,900 firms retain valid TSUE classifications.

3.3 Event definitions

We test two event date definitions to assess sensitivity. The *fiscal year-end* (FYE) event date is December 31, 2023, for all firms—the date at which the fiscal year's economic activity concludes. The *announcement date* (Ann) event is the actual earnings announcement date reported in I/B/E/S (`anndats_act`), which varies across firms and falls predominantly in the first quarter of 2024. The FYE definition follows Ball and Brown [1968] more closely, while the announcement date definition better captures the market's reaction to the information release itself.

3.4 Abnormal return models

We compute abnormal returns using five specifications of increasing complexity, spanning the models most commonly used in the event study literature [Kothari and Warner, 2007].

Models 1–2: Simple market-adjusted returns. Following the exact methodology of Ball and Brown [1968], abnormal returns are computed as

$$AR_{i,t}^{\text{Simple}} = R_{i,t} - R_{m,t}, \tag{4}$$

where $R_{m,t}$ is either the CRSP equal-weighted market return (Model 1) or the CRSP value-weighted market return (Model 2). This approach assumes $\alpha_i = 0$ and $\beta_i = 1$ for all firms.

Models 3–4: Regression-based market model. Following Brown and Warner [1985], we estimate the market model

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t} \tag{5}$$

over an estimation window of months -24 to -4 relative to the event (requiring a minimum of 12 monthly observations), using either the equal-weighted (Model 3) or value-weighted (Model 4) market return as the benchmark. Abnormal returns are $AR_{i,t}^{\text{MM}} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{m,t})$.

Model 5: Fama–French–Carhart four-factor model. We estimate

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i \text{MKT}_t + s_i \text{SMB}_t + h_i \text{HML}_t + p_i \text{UMD}_t + \varepsilon_{i,t} \tag{6}$$

over the same estimation window, where MKT, SMB, HML, and UMD are the market, size, value, and momentum factors from the Kenneth French Data Library [Fama and French, 1993, Carhart, 1997]. Abnormal returns are

$$AR_{i,t}^{\text{FFC}} = (R_{i,t} - R_{f,t}) - (\hat{\alpha}_i + \hat{\beta}_i \text{MKT}_t + \hat{s}_i \text{SMB}_t + \hat{h}_i \text{HML}_t + \hat{p}_i \text{UMD}_t). \tag{7}$$

3.5 Abnormal Performance Index

Following Ball and Brown [1968], we aggregate abnormal returns using the multiplicative Abnormal Performance Index (API). For each event month τ within the window, we first compute the cross-sectional average abnormal return:

$$\overline{AR}_\tau = \frac{1}{N_\tau} \sum_{i=1}^{N_\tau} AR_{i,\tau}, \tag{8}$$

where N_τ is the number of firms with valid abnormal returns in month τ. The API is then

$$\text{API}_\tau = \prod_{s=-12}^{\tau} (1 + \overline{AR}_s), \tag{9}$$

normalized to 1.0 at the start of the event window ($\tau = -12$). We compute separate API trajectories for good-news and bad-news portfolios. The spread at any month τ is defined as $\text{API}_\tau^{\text{Good}} - \text{API}_\tau^{\text{Bad}}$.

3.6 Statistical inference

We assess the statistical significance of the good-news/bad-news spread using bootstrap confidence intervals based on firm-level block resampling [Petersen, 2009]. In each of 1,000 bootstrap iterations, we draw firms with replacement (preserving each firm's complete time series), recompute the API for both portfolios, and record the spread at month zero. The 95% confidence interval is constructed from the 2.5th and 97.5th percentiles of the bootstrap distribution.

For cross-sectional regressions, we estimate the relationship between cumulative abnormal returns (CAR) over the three-month window $[-1, +1]$ surrounding the announcement date and the SUE measure, controlling for firm characteristics. The specification is

$$\text{CAR}_{i,[-1,+1]} = \gamma_0 + \gamma_1 \text{SUE}_i + \boldsymbol{\gamma}_2' \mathbf{X}_i + \delta_j + u_i, \tag{10}$$

where $\mathbf{X}_i$ includes the natural logarithm of market capitalization [Banz, 1981, Fama and French, 1992], the book-to-market ratio [Fama and French, 1992], prior 12-month momentum returns [Jegadeesh and Titman, 1993], the natural logarithm of analyst coverage [Hong et al., 2000], analyst forecast dispersion as a proxy for earnings volatility [Foster et al., 1984], and financial leverage. The term δ_j denotes two-digit SIC industry fixed effects. All continuous control variables are winsorized at the 1st and 99th percentiles. Standard errors are heteroskedasticity-robust (HC1) following White [1980].

3.7 Data sources

Monthly stock returns are from the CRSP Monthly Stock File (`crsp.msf`). Market returns (equal-weighted and value-weighted) are from the CRSP Monthly Stock Index File (`crsp.msi`). Fama–French–Carhart factor returns are from the Kenneth French Data Library via WRDS (`ff.factors_monthly`). Analyst forecast data, actual earnings, and announcement dates are from I/B/E/S (`ibes.statsum_epsus`). Firm fundamentals for control variables (total assets, common equity, total liabilities, share price, shares outstanding, SIC codes) are from Compustat (`comp.funda`), linked to CRSP via the CCM linking table (`crsp.ccmxpf_linktable`). All returns are winsorized at the 1st and 99th percentiles.

4 Empirical results

4.1 The attenuation pattern across models

Table 2 presents the central results of our decomposition: API values for good-news and bad-news portfolios at the event date (month zero) across all 20 specifications. Three patterns emerge.

First, the good-news/bad-news spread is positive in 18 of 20 specifications, confirming the information content of earnings (**H1**). The two exceptions—FFC with TSUE under both event definitions—show spreads of essentially zero (0.16 and −0.24 percentage points),

consistent with the time-series surprise signal being fully absorbed by factor adjustment.

Second, the spread attenuates monotonically as the abnormal return model incorporates additional risk factors (**H2**). Under the fiscal year-end event with TSUE (the specification closest to Ball and Brown's original methodology), the spread is 12.98 percentage points for simple equal-weighted adjustment, 12.20 for simple value-weighted, 3.64 for the regression-based equal-weighted market model, 3.61 for the value-weighted market model, and 0.16 for the Fama–French–Carhart four-factor model. This represents a decline of nearly 99% from the simplest to the most comprehensive specification. Even with SUE, which generates smaller initial spreads, the attenuation from simple equal-weighted (9.00 percentage points using announcement dates) to four-factor (3.15 percentage points) represents a 65% reduction.

Third, the choice of surprise measure matters substantially. TSUE produces larger spreads than SUE under simple models (e.g., 12.98 versus 6.96 percentage points for FYE with equal-weighted adjustment) but smaller or negligible spreads under factor-adjusted models. This divergence suggests that the TSUE classification correlates with systematic factor exposures—particularly momentum and value—that are removed by the four-factor model, while the SUE classification captures genuinely idiosyncratic information.

Figure 1 displays the API trajectories for the simple equal-weighted specification across all four combinations of event definition and surprise measure. The characteristic Ball and Brown pattern—a gradual divergence between good-news and bad-news portfolios beginning approximately 12 months before the event—is clearly visible. Figure 2 illustrates the attenuation pattern for the announcement date with SUE, showing the progressive compression of the spread as models add risk factors.

4.2 Bootstrap confidence intervals

Table 3 reports bootstrap confidence intervals for six key specifications, constructed from 1,000 firm-level block resamples. All simple market-adjusted specifications produce spreads

with confidence intervals that exclude zero, confirming statistical significance. The Ball and Brown exact specification (Simple EW + TSUE + FYE) yields a mean bootstrap spread of 13.00 percentage points with a 95% confidence interval of $[10.42, 15.66]$. The four-factor model with SUE and announcement dates (FFC + SUE + Ann) remains marginally significant, with a mean spread of 3.16 and a confidence interval of $[0.55, 5.80]$ that excludes zero. The four-factor model with TSUE (FFC + TSUE + Ann), however, yields a mean spread of -0.26 percentage points with a confidence interval of $[-2.57, 1.90]$ that includes zero, confirming that the time-series surprise signal is fully absorbed by factor adjustment.

4.3 Quintile monotonicity

To verify that the earnings–return association is not driven by extreme observations, we sort firms into quintiles based on SUE and compute the API for each quintile portfolio using simple equal-weighted market-adjusted returns with the fiscal year-end event. Table 4 reports the results. The API at month zero increases monotonically from Q2 through Q5, with Q5 (most positive surprise) reaching 1.1755 and Q1 (most negative surprise) at 1.0818. The one departure from strict monotonicity is between Q1 and Q2, where Q1 slightly exceeds Q2 (1.0818 versus 1.0576); this likely reflects a small number of extreme negative surprises in Q1 that experienced mean reversion during FY2023. The Spearman rank correlation between quintile rank and API at month zero is 0.90 ($p = 0.037$), confirming a significant monotonic relationship. Figure 3 displays the full API trajectories for all five quintile portfolios, showing the characteristic fan pattern with the spread widening progressively over the event window.

4.4 Cross-sectional evidence

Table 5 presents cross-sectional regressions of cumulative abnormal returns around the announcement date (CAR$[-1, +1]$) on SUE and control variables. We estimate three nested specifications for each of three abnormal return definitions: (1) univariate with SUE only; (2) adding size, book-to-market, and momentum; and (3) the full model with analyst cover-

age, earnings volatility, leverage, and two-digit SIC industry fixed effects.

The coefficient on SUE is positive and statistically significant across all nine specifications. In the univariate regression using simple equal-weighted abnormal returns, a one-unit increase in SUE is associated with a 0.47 percentage point increase in CAR ($t = 5.44$, $p < 0.001$). Adding the full set of controls and industry fixed effects reduces the coefficient to 0.30 percentage points but it remains highly significant ($t = 3.97$, $p < 0.001$). For the Fama–French–Carhart four-factor abnormal returns, the univariate coefficient is 0.32 percentage points ($t = 2.81$, $p = 0.005$) and the full specification yields 0.25 percentage points ($t = 2.43$, $p = 0.015$). The persistence of significance across all specifications and abnormal return definitions confirms that SUE carries genuine predictive power for announcement-window returns that is not subsumed by known cross-sectional return predictors or industry effects.

The adjusted R^2 increases substantially from the univariate specification (1.4% for simple EW) to the full specification (27.0%), indicating that the control variables explain considerable variation in announcement returns. Importantly, the SUE coefficient attenuates only modestly (from 0.47 to 0.30 percentage points) despite the large increase in explanatory power, suggesting that the earnings surprise signal is largely orthogonal to the control variables.

4.5 Post-earnings announcement drift

Table 6 reports the evolution of the good-news/bad-news spread from the announcement date through month +12 for three model specifications. Under simple equal-weighted adjustment, the spread widens from 9.00 percentage points at month zero to 17.79 at month +3, stabilizing thereafter at approximately 17–19 percentage points, reaching 18.97 at month +12. This represents a post-announcement drift of 9.98 percentage points—nearly doubling the initial spread.

The drift is robust to risk adjustment. Under the value-weighted market model, the

spread increases from 5.23 to 10.57 percentage points (drift of 5.33), and under the Fama–French–Carhart four-factor model, from 3.15 to 7.30 percentage points (drift of 4.15). The persistence of drift under four-factor adjustment is particularly notable: it implies that the market's incomplete initial response to earnings news cannot be attributed to compensation for market, size, value, or momentum risk. Figure 4 displays the API trajectories over the full $[-12, +12]$ window for all three specifications, with the post-announcement region highlighted.

4.6 Size-conditional PEAD

Table 7 reports spreads at month zero and month +12 separately for small-capitalization and large-capitalization firms, split at the median market capitalization (median ln(size) = 7.98). At month zero, the initial spread is similar across size groups: 7.36 percentage points for small caps versus 8.12 for large caps under simple equal-weighted adjustment, and 5.67 versus 4.08 under the Fama–French–Carhart model.

The divergence emerges dramatically in the post-announcement period, consistent with **H3**. Small-cap firms exhibit a post-announcement drift of 14.25 percentage points (spread widening from 7.36 to 21.61), compared with only 3.03 percentage points for large caps (from 8.12 to 11.15). This nearly fivefold difference supports the limited-attention hypothesis of Hong et al. [2000]: earnings information diffuses slowly into the prices of smaller firms that attract less analyst coverage and institutional investor attention. Figure 5 visualizes this interaction, showing the marked divergence between small-cap and large-cap drift trajectories in the post-announcement period.

5 Discussion

Our results speak to three interconnected themes: market efficiency, the role of risk adjustment in interpreting event study evidence, and the practical implications for portfolio

construction.

5.1 Market efficiency

The evidence presents a nuanced picture of market efficiency that varies by time horizon. In the pre-announcement period, our findings strongly support semi-strong form efficiency as defined by Fama [1970]. Across all 20 specifications, good-news and bad-news portfolios begin diverging approximately 12 months before the event, with the bulk of the divergence occurring well before the formal earnings release. This anticipation—visible in Figures 1 and 3—confirms Ball and Brown's original finding that the market incorporates earnings-relevant information from quarterly reports, management guidance, analyst revisions, industry conditions, and macroeconomic developments throughout the fiscal year.

At the announcement date, however, the evidence is mixed. The significant spreads documented in Table 2 indicate that the announcement itself conveys incremental information not yet reflected in prices. Under four-factor adjustment with analyst-based surprises, this residual information content is approximately 3 percentage points—economically meaningful but substantially smaller than the 9–14 percentage points measured under simple market adjustment. The attenuation pattern documented in Figure 2 underscores the importance of Fama's (1970) joint hypothesis: much of what appears to be "information content" under simple models is in fact compensation for systematic risk exposures.

In the post-announcement period, the continued drift documented in Table 6 and Figure 4 challenges semi-strong efficiency. The spread approximately doubles from month zero to month +12 across all model specifications, and this drift survives Fama–French–Carhart four-factor adjustment (from 3.15 to 7.30 percentage points). Following Fama [1998], one might argue that this drift reflects compensation for an omitted risk factor. However, the concentration of drift in small-capitalization stocks (Table 7 and Figure 5) is more consistent with the behavioral explanation of Hong et al. [2000]: information diffuses slowly into the prices of firms that attract limited investor attention, generating predictable post-

announcement returns that represent genuine inefficiency rather than risk compensation.

5.2 Implications for event study methodology

Our systematic decomposition carries an important methodological implication. The measured "information content" of any corporate event depends critically on the abnormal return model employed. In our sample, the simple equal-weighted market-adjusted model—which remains widely used due to its simplicity and direct connection to Ball and Brown [1968]—produces a spread roughly three times larger than the Fama–French–Carhart four-factor model for analyst-based surprises (9.00 versus 3.15 percentage points at the announcement date). Researchers using simple market adjustment to evaluate the information content of earnings announcements, management forecasts, analyst revisions, or other corporate events may systematically overstate the magnitude of the market's response to firm-specific information.

This finding echoes the concern raised by Fama [1998] about the sensitivity of anomaly evidence to methodology, and extends it to the event study setting. We recommend that future event studies report results under multiple model specifications—at minimum, both simple market-adjusted and four-factor-adjusted returns—to provide readers with a transparent assessment of how much measured abnormal performance is attributable to factor exposures versus genuine information content.

The divergence between SUE and TSUE under factor adjustment also carries methodological implications. Time-series surprise measures, while faithful to the original Ball and Brown [1968] methodology, appear to classify firms in ways that correlate with systematic factors. Analyst consensus forecasts, by contrast, provide a cleaner separation between firm-specific news and factor-related return variation. For researchers seeking to isolate the market's response to idiosyncratic earnings information, analyst-based surprise measures are preferable.

5.3 Implications for portfolio management

For practitioners, our findings offer several actionable insights. First, earnings-based trading strategies generate genuine alpha after comprehensive risk adjustment, but the magnitude is considerably smaller than naive event study estimates suggest. A portfolio manager evaluating an earnings surprise strategy should benchmark against a four-factor model, not a simple market index, to obtain a realistic assessment of expected alpha.

Second, the alpha concentrates in small-capitalization stocks. The post-announcement drift for small caps is 14.25 percentage points over 12 months, compared with only 3.03 for large caps. This implies that earnings-based strategies are most profitable in the small-cap segment of the market—precisely where implementation costs (bid-ask spreads, market impact) are highest. Whether the net-of-transaction-cost alpha remains positive for small caps is an important question that our analysis does not address but that portfolio managers should evaluate before deploying capital.

Third, the timing of alpha accrual matters. Table 6 shows that the majority of post-announcement drift occurs in the first three months: the spread jumps from 9.00 to 17.79 percentage points between months zero and +3 under simple adjustment, with more modest increments thereafter. This suggests that earnings-based strategies should focus on the $[0, +3]$ month window to capture the bulk of the drift, rather than holding positions for the full 12-month post-announcement period.

Fourth, our finding that TSUE-based strategies offer zero alpha after factor adjustment (Table 3: spread $= -0.26$, 95% CI $[-2.57, 1.90]$) has a direct practical implication: strategies based on time-series earnings models are unlikely to generate risk-adjusted returns. The apparent profitability of such strategies under simple benchmarks is an artifact of factor loading, not genuine information content.

6 Conclusion

This study provides a systematic decomposition of the earnings information content first documented by Ball and Brown [1968], quantifying how the measured good-news/bad-news portfolio spread changes as the abnormal return model incorporates progressively more systematic risk factors. Using 2,358 U.S. firms with December 2023 fiscal year-ends and testing 20 specifications that vary the abnormal return model, earnings surprise measure, and event date definition, we document three principal findings.

First, the information content of earnings persists in modern data. The spread between good-news and bad-news portfolios is positive and statistically significant in 18 of 20 specifications, and the Ball and Brown exact methodology produces a spread of 12.98 percentage points—closely matching the original result from over half a century ago.

Second, the bulk of traditionally measured information content is attributable to compensation for systematic risk factors. Under time-series surprises, the spread attenuates by 99%, from 12.98 percentage points to 0.16 percentage points, as the model progresses from simple market adjustment to Fama–French–Carhart four-factor adjustment. Under analyst-based surprises, the attenuation is 65%, from 9.00 to 3.15 percentage points. Analyst-based earnings surprises retain significant information content after factor adjustment; time-series surprises do not.

Third, post-earnings announcement drift remains economically large and concentrates in small-capitalization stocks. The spread approximately doubles from the announcement date to month +12, and the drift is nearly five times larger for small caps (14.25 percentage points) than for large caps (3.03 percentage points), consistent with the limited-attention hypothesis of Hong et al. [2000].

Our study is subject to several limitations. The analysis focuses on a single fiscal year (FY2023) and on firms with December fiscal year-ends, which may limit generalizability across time periods and fiscal year-end months. The monthly return frequency, while consistent with Ball and Brown [1968], may obscure finer dynamics visible at daily frequency. We

do not account for transaction costs, which are particularly relevant for the small-cap PEAD strategy where implementation frictions are substantial. Finally, our cross-sectional analysis employs a single-period design; a panel approach spanning multiple fiscal years would provide greater statistical power and allow for time-varying coefficient estimation.

These limitations suggest natural extensions. A multi-year panel study could assess whether the attenuation pattern and size-conditional drift are stable features of the data or specific to the FY2023 period. International evidence would reveal whether factor compensation explains a similar share of earnings information content in markets with different institutional structures. Daily-frequency analysis could sharpen the identification of the announcement response and separate the announcement effect from the drift more precisely. We leave these extensions for future research.

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Table 2: Abnormal Performance Index at month zero: all 20 specifications

Event	Model	Surprise	Good API	Bad API	Spread (pp)
FYE	Simple (EW)	SUE	1.1353	1.0658	6.96**
	Simple (EW)	TSUE	1.1582	1.0283	12.98***
	Simple (VW)	SUE	1.0191	0.9539	6.52**
	Simple (VW)	TSUE	1.0427	0.9206	12.20***
	MM (EW)	SUE	0.9927	0.9490	4.38*
	MM (EW)	TSUE	0.9595	0.9230	3.64*
	MM (VW)	SUE	0.9924	0.9552	3.71*
	MM (VW)	TSUE	0.9579	0.9218	3.61*
	FFC	SUE	1.0082	0.9811	2.71*
	FFC	TSUE	0.9693	0.9676	0.16
Ann	Simple (EW)	SUE	1.1417	1.0517	9.00**
	Simple (EW)	TSUE	1.1711	1.0301	14.09***
	Simple (VW)	SUE	0.9566	0.8806	7.60**
	Simple (VW)	TSUE	0.9807	0.8664	11.44***
	MM (EW)	SUE	1.0239	0.9592	6.47**
	MM (EW)	TSUE	0.9912	0.9557	3.54*
	MM (VW)	SUE	0.9339	0.8816	5.23**
	MM (VW)	TSUE	0.9027	0.8800	2.27*
	FFC	SUE	1.0063	0.9747	3.15*
	FFC	TSUE	0.9608	0.9632	−0.24

This table reports the Abnormal Performance Index (API) at event month zero for good-news and bad-news portfolios across 20 specifications. The API is computed using the multiplicative formulation of Ball and Brown [1968], normalized to 1.0 at month −12. Spread is the difference between good-news and bad-news API, expressed in percentage points. FYE denotes fiscal year-end (December 31, 2023); Ann denotes the actual earnings announcement date. Simple (EW) and Simple (VW) are market-adjusted returns using equal-weighted and value-weighted CRSP market returns, respectively. MM denotes the regression-based market model. FFC denotes the Fama–French–Carhart four-factor model. SUE is analyst-based Standardized Unexpected Earnings; TSUE is time-series unexpected earnings. Significance based on the magnitude of the spread: *** > 10pp, ** > 5pp, * > 2pp.

Table 3: Bootstrap confidence intervals for good-news/bad-news spread at month zero

Specification	Mean spread (pp)	95% CI	Significant
Simple EW + TSUE + FYE (B&B exact)	13.00	$[10.42, 15.66]$	Yes
Simple EW + SUE + FYE	7.00	$[4.14, 9.63]$	Yes
Simple EW + SUE + Ann	9.00	$[6.24, 11.81]$	Yes
MM VW + SUE + Ann	5.27	$[2.81, 7.53]$	Yes
FFC + SUE + Ann	3.16	$[0.55, 5.80]$	Yes
FFC + TSUE + Ann	-0.26	$[-2.57, 1.90]$	No

Bootstrap confidence intervals are constructed from 1,000 iterations of firm-level block resampling. In each iteration, firms are drawn with replacement (preserving each firm's complete return time series), and the API spread between good-news and bad-news portfolios is computed at month zero. The 95% confidence interval is the $[2.5\text{th}, 97.5\text{th}]$ percentile of the bootstrap distribution. A specification is significant if the confidence interval excludes zero.

Table 4: SUE quintile portfolios: API at month zero

Quintile	N	API(0)	Cumulative AR (%)
Q1 (most negative SUE)	460	1.0818	8.18
Q2	484	1.0576	5.76
Q3	435	1.0962	9.62
Q4	459	1.1328	13.28
Q5 (most positive SUE)	460	1.1755	17.55
Q5 − Q1 spread			9.37

Firms are sorted into quintiles based on SUE. The API is computed using simple equal-weighted market-adjusted returns with the fiscal year-end event date, normalized to 1.0 at month -12. Cumulative AR is $(\text{API}(0) - 1) \times 100$. Spearman rank correlation between quintile and API(0): $\rho = 0.90$ ($p = 0.037$).

Table 5: Cross-sectional regressions: CAR$[-1, +1]$ on SUE

Dep. variable	Specification	$\hat{\gamma}_{\text{SUE}}$	t-stat	p-value	R^2	N
CAR (Simple EW)	(1) Univariate	0.00474	5.444	0.000	0.014	2,306
	(2) +Size, BM, Mom	0.00306	4.176	0.000	0.177	2,279
	(3) Full + Ind. FE	0.00297	3.972	0.000	0.270	2,277
CAR (MM VW)	(1) Univariate	0.00433	4.503	0.000	0.009	2,306
	(2) +Size, BM, Mom	0.00315	3.811	0.000	0.142	2,279
	(3) Full + Ind. FE	0.00349	4.109	0.000	0.281	2,277
CAR (FFC)	(1) Univariate	0.00322	2.807	0.005	0.005	2,306
	(2) +Size, BM, Mom	0.00245	2.430	0.015	0.120	2,279
	(3) Full + Ind. FE	0.00254	2.428	0.015	0.249	2,277

The dependent variable is the cumulative abnormal return over months $[-1, +1]$ surrounding the announcement date. Specification (1) includes only SUE. Specification (2) adds the natural logarithm of market capitalization, book-to-market ratio, and prior 12-month momentum return. Specification (3) further adds the natural logarithm of analyst coverage, analyst forecast dispersion (earnings volatility), financial leverage, and two-digit SIC industry fixed effects. Standard errors are heteroskedasticity-robust (HC1) following White [1980]. All continuous variables are winsorized at the 1st and 99th percentiles.

Table 6: Post-earnings announcement drift: spread evolution

	Spread (pp) at month					Post-ann.
Model	0	+3	+6	+9	+12	drift
Simple (EW) + SUE	9.00	17.79	16.89	18.14	18.97	+9.98
MM (VW) + SUE	5.23	11.63	10.08	9.95	10.57	+5.33
FFC + SUE	3.15	9.66	7.47	7.47	7.30	+4.15

This table reports the good-news minus bad-news API spread (in percentage points) at selected months following the announcement date. Post-announcement drift is the change in spread from month 0 to month +12. All specifications use analyst-based SUE for portfolio classification. The API is normalized to 1.0 at month -12 and computed using the extended event window $[-12, +12]$.

Table 7: Size-conditional spreads and post-earnings announcement drift

Model	Size group	N firms		Spread(0)	Spread(+12)	Drift
		Good	Bad			
Simple (EW)	Small	588	492	7.36	21.61	+14.25
	Large	758	342	8.12	11.15	+3.03
FFC	Small	588	492	5.67	—	—
	Large	758	342	4.08	—	—

Firms are split into small and large groups at the median market capitalization (ln(size) = 7.98). Spreads are reported in percentage points. Drift is the change in spread from month 0 to month +12. PEAD by size is computed using simple equal-weighted market-adjusted returns with the announcement date event and SUE classification. The FFC drift by size is reported only for the main window $[-12, +6]$; the extended window analysis focuses on simple market-adjusted returns to maintain sample size within subgroups.

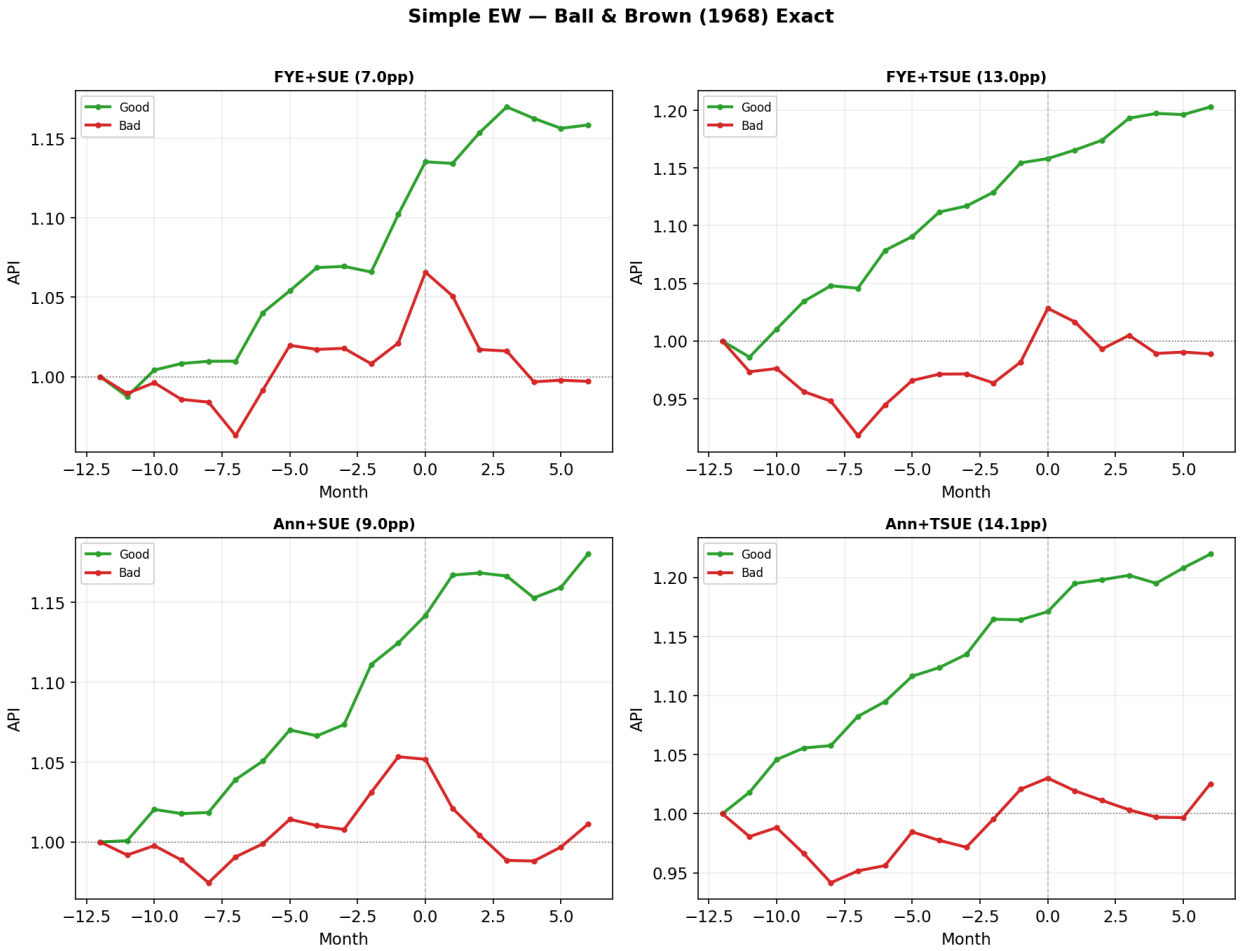

Figure 1: Simple equal-weighted market-adjusted API trajectories. Each panel shows the Abnormal Performance Index for good-news and bad-news portfolios over event months −12 to +6, using the simple equal-weighted market-adjusted abnormal return specification of Ball and Brown [1968]. Top left: FYE + SUE (spread 7.0pp). Top right: FYE + TSUE (spread 13.0pp). Bottom left: Ann + SUE (spread 9.0pp). Bottom right: Ann + TSUE (spread 14.1pp).

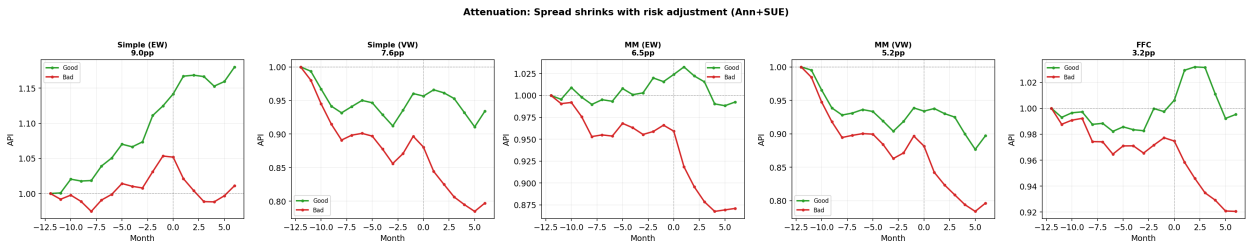

Figure 2: Model comparison: attenuation of the good-news/bad-news spread (Ann + SUE). From left to right: Simple (EW), Simple (VW), MM (EW), MM (VW), and FFC. The spread at month zero declines monotonically from 9.0pp to 3.15pp as risk adjustment becomes more comprehensive.

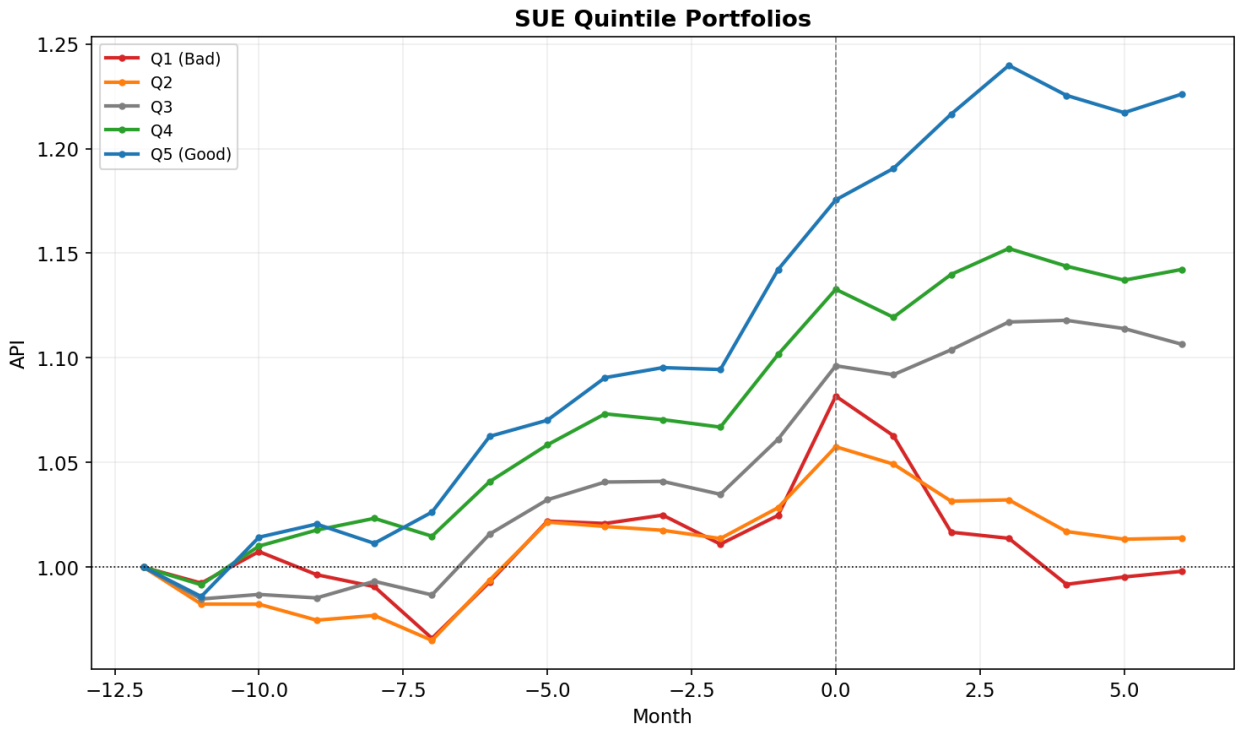

Figure 3: SUE quintile portfolio API trajectories. Firms are sorted into quintiles based on SUE magnitude. The API is computed using simple equal-weighted market-adjusted returns with the fiscal year-end event. The fan pattern confirms a monotonic association between earnings surprise magnitude and cumulative abnormal performance.

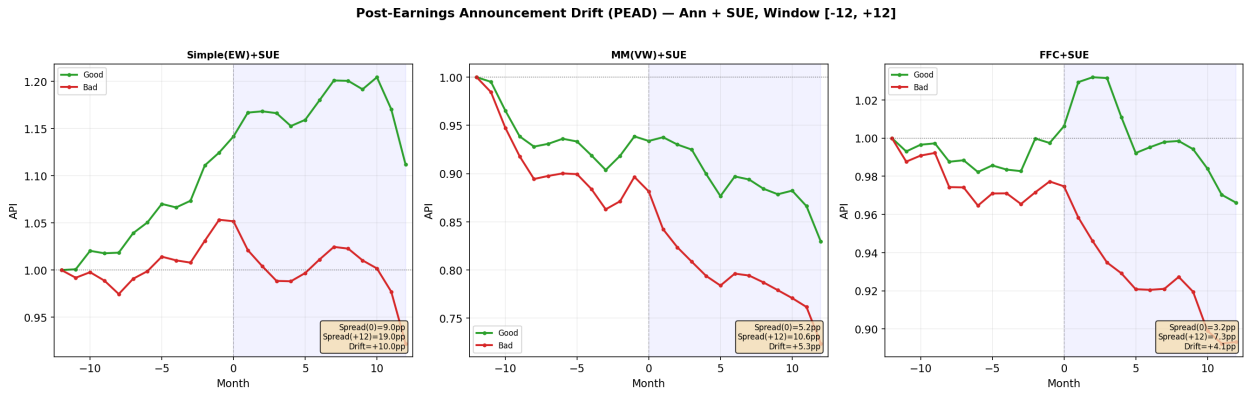

Figure 4: Post-earnings announcement drift (PEAD): API trajectories over the extended $[-12, +12]$ month window (Ann + SUE). Left: Simple (EW). Center: MM (VW). Right: FFC. The shaded region indicates the post-announcement period. Annotations report the spread at month 0, month +12, and the post-announcement drift.

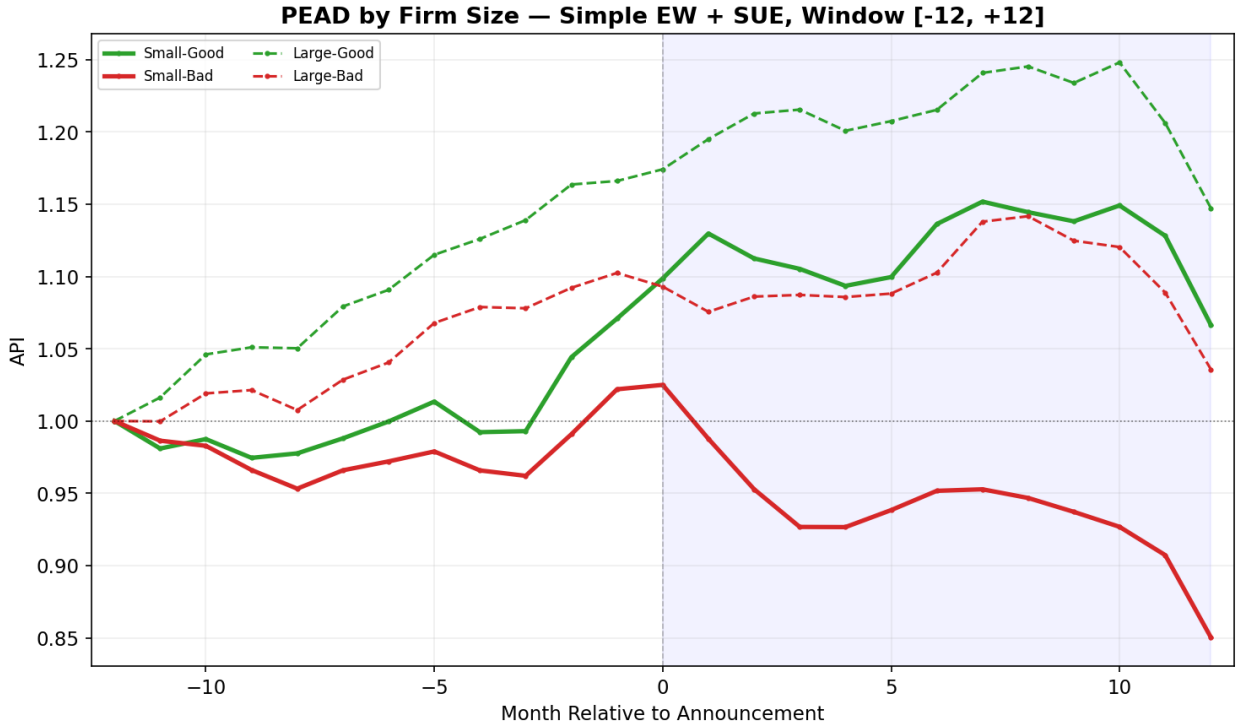

Figure 5: PEAD by firm size: Simple EW + SUE, window $[-12, +12]$. Solid lines represent small-capitalization firms; dashed lines represent large-capitalization firms. Green indicates good news; red indicates bad news. Small-cap drift is 14.25pp versus 3.03pp for large caps, consistent with the limited-attention hypothesis.